Practice Exam Numerical Recipes for Astrophysics A

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Solutions

1) Knowledge (1 point each)

(a) A new point in parameter space is chosen based on the current point, in a way that satisfies detailed balance (so that we sample ergodically). (This is done by accepting the new point with a probability that depends on the ratio of the likelihoods at the current and the next point.) It is used in the Markov Chain Monte Carlo method.

(b) A closed method evaluates the integrand at the end points (limits) of the integral, while an open method does not (useful when the integrand cannot be numerically evaluated there). A semi-open method only evaluates the integrand at one of the limits.

(c) $k + 1$ points.

(d) It's swapping matrix rows so that the first non-zero element of each row is on the diagonal; or, more generally, it is about swapping elements to get desirable elements in certain positions.

(e) $|x - b| \gtrsim \sqrt{\epsilon_{\mathrm{m}}} b$ (also full points for an equal sign), where we've used that the machine error is a fractional/relative error.

2) Understanding (2 points each)

(a) `15` in binary is a bunch of zeroes ending with four ones. The `&` (AND) operation therefore selects only those bits of `n` corresponding to numbers below 16, meaning it is equivalent to `n mod 16`.

(b) The second is more efficient, because it minimizes the number of multiplications needed.

(c) One example is the $l1$ estimator, which weights by the absolute distance between data and model rather than its square, like least-squares and χ^2 do. This way outliers receive less weight and cannot skew the function as much.

(d) For a simple method, the error scales as the number of evaluations to some negative power (e.g. $\mathcal{O}(N^{-2})$ for extended trapezoid, or $\mathcal{O}(N^{-4})$ for extended Simpson's). Romberg integration, on the other hand, combines approximations to the integral to remove the leading error term, thus effectively decreasing the power, effectively giving exponential convergence.

(e) Quicksort can be very slow when the pivot is near the lowest or highest element, so you might prefer something with better worse-case behaviour. Mergesort is also stable while quicksort is not. Finally, you might just prefer a different one because of how it works/how easily you can implement it, since they're typically all about equally good anyway.

(f) The 64-bit XOR shift shifts the bits of an unsigned 64-bit integer x left or right by some varying number of positions, then does a bit-wise exclusive OR (XOR) operation between the result and x itself. It then repeats this twice, alternating the direction of the shift. This is better than the LCG because the number sequence is far less correlated with itself, even though both can have an equal period and no repeated occurrences.

(g) We first transform to the range $[0, 1)$ by dividing by 2^{32}, then multiply by $12 - 5 = 7$, and finally add 5.

(h) Say the equations we were solving were $\mathbf{Ax} = \mathbf{b}$. If we have the LU decomposition of $\mathbf{A}$, we can estimate the error on $\mathbf{x}$ and subtract it. Calling the found approximation to the solution $\mathbf{x}'$, we can calculate $\mathbf{Ax}' = \mathbf{b}'$, with known error $\mathbf{b}' - \mathbf{b}$. The unknown error $\mathbf{x}' - \mathbf{x} \equiv \boldsymbol{\delta}\mathbf{x}$ can then be estimated by solving $\mathbf{A}\boldsymbol{\delta}\mathbf{x} = \mathbf{b}' - \mathbf{b}$, and a better solution $\mathbf{x}'' = \mathbf{x}' - \boldsymbol{\delta}\mathbf{x}$ can be calculated. This process can be repeated and is fast, since the LU decomposition doesn't have to be recalculated.

(i) x is the particular value of χ^2 we found, $k = N - M$ is the number of degrees of freedom. $p(x, k)$ gives the probability of observing a particular value of χ^2, given the model is correct. By calculating a p-value as $Q = 1 - P$, with $P(y < x, k)$ the cumulative distribution function (CDF) of $p(x, k)$, we can determine a goodness-of-fit for our best-fit model.

(j) A cubic spline is a piecewise cubic interpolation, where the third-order polynomial in every interval has matching values, derivatives and second derivatives with that of its neighbour at the interval edges. This gives $3N - 5$ constraints for $3N - 3$ unknown coefficients. The "natural" part means that we choose the last two constraints to be that the second derivatives at the endpoints are zero.

3) Application (3 points each)

(a) There are two (or three) problems: the integral is improper since one of its limits goes to infinity, and the integrand goes to infinity at $z = 0$. A possible third problem is the fact that the integrand is not analytical due to the absolute value. We should first split the integral at $z = 0$:

$$\int_{-5}^{0} (-z)^{-1/2} e^{-z/2} \, \mathrm{d}z + \int_{0}^{\infty} z^{-1/2} e^{-z/2} \, \mathrm{d}z.$$

The first part can now be solved by an open method such as midpoint. For the second part, we need a transformation to get rid of ∞, such as $x \equiv e^{-z/2}$ which gives:

$$\int_{0}^{1} \frac{2}{\sqrt{-2 \ln x}} \, \mathrm{d}x.$$

We again need an open method to solve this integral.

(b) The values at interior points are assumed to evolve linearly between the grid points, in the horizontal, vertical and diagonal directions. We can write this as:

$$f(x, y) = a + b(x - x_0) + c(y - y_0) + d(x - x_0)(y - y_0).$$

Since we know the values at the grid points, we can fill these in and solve for a, b, c and d. We find $a = f(x_0, y_0)$, $b = [f(x_1, y_0) - f(x_0, y_0)]/[x_1 - x_0]$, $c = [f(x_0, y_1) - f(x_0, y_0)]/[y_1 - y_0]$ and $d = [f(x_1, y_1) - f(x_1, y_0) - f(x_0, y_1) + f(x_0, y_0)]/[(x_1 - x_0)(y_1 - y_0)]$. Alternatively, we could have defined $s \equiv (x - x_0)/(x_1 - x_0)$ and $t \equiv (x - x_0)/(x_1 - x_0)$, and write:

$$f(x, y) = (1 - s)(1 - t)f(x_0, y_0) + s(1 - t)f(x_1, y_0) + (1 - s)tf(x_0, y_1) + stf(x_1, y_1).$$

(c) How the solution looks exactly depends on the pivot choice. The most important thing is that the pivot is chosen consistently. Here I'll do it like in the lecture: first pivot is the median of the first, last and middle, all subsequent will be the middle element rounded up. At every step, elements are sorted relative to the pivot, by comparing two iterators that go from left to right and vice versa, comparing elements to the pivot. After the iterators cross, the two subarrays on either side of the pivot get sorted recursively (the pivot is in its final position at that point). We get:

$$|5|7|3|2|4|9|1| \to |5|7|3|\mathbf{2}|4|9|1| \to |1|\mathbf{2}|3|7|4|9|5| \to |1|2| \quad |3|7|\mathbf{4}|9|5| \to$$

$$|1|2| \quad |3|\mathbf{4}|7|9|5| \to |1|2| \quad |3|4| \quad |7|\mathbf{9}|5| \to |1|2| \quad |3|4| \quad |7|5|\mathbf{9}| \to$$

$$|1|2| \quad |3|4| \quad |7|\mathbf{5}| \quad |9| \to |1|2| \quad |3|4| \quad |\mathbf{5}|7| \quad |9| \to |1|2|3|4|5|7|9|$$

(d) From the initial simplex, the method reflects the worst (highest) point and expands it (1), reflects three times (2, 3, 4), then contracts (5). At that point the contracted point is now the best point, and we return it as the minimum (shown in red in the figure below). It doesn't matter if the students did not find the exact same steps or solution, what matter is that at each step they made the right choice for the result they found and applied it correctly.

(e) The method is Monte Carlo integration. First we need to scale these points so that they lie in a rectangular volume (A_{box}) around the shaded region (A_{shaded}). Easiest is to scale x to $1 + 4x$ and y to $10 + 10y$, but tighter ranges across the area to integrate are of course fine too. Doing so reveals that six of the 20 pairs fall inside the red area, which leads to the area estimate $A_{\text{shaded}} \approx (N_{\text{inside}}/N) \times A_{\text{box}} = (6/20) \times (4 \times 10) = 12$. The true area is about 11.0298, so this is a pretty good estimate, given the low number of points (in fact, it is as close as we could have gotten, so we got lucky).

Bonus points on an exam are deliberately not easy to obtain; we did not discuss the uncertainty of MC integration, so you need to reason your way there. The correct standard deviation is about 4.0, but you are unlikely to recover this exactly – nor do you need to. One way to derive an uncertainty is to realize that we can view this problem as a mean-value problem, where we are calculating the mean of points inside (carrying weight 1) and outside the shaded region (carrying weight 0). The error on the area is then $\sigma = A_{\text{box}} \times \sigma_{\text{frac}}/\sqrt{N}$. This involves calculating σ_{frac}, the inherent scatter in landing inside the area or not, which for $f = A_{\text{shaded}}/A_{\text{box}}$ is $\sigma_{\text{frac}} = \sqrt{[f \times N \times (1 - f)^2 + (1 - f) \times N \times (0 - f)^2]/N} = \sqrt{f(1 - f)}$ (if we do not know f independently, as we do not here, notice that we actually have to divide

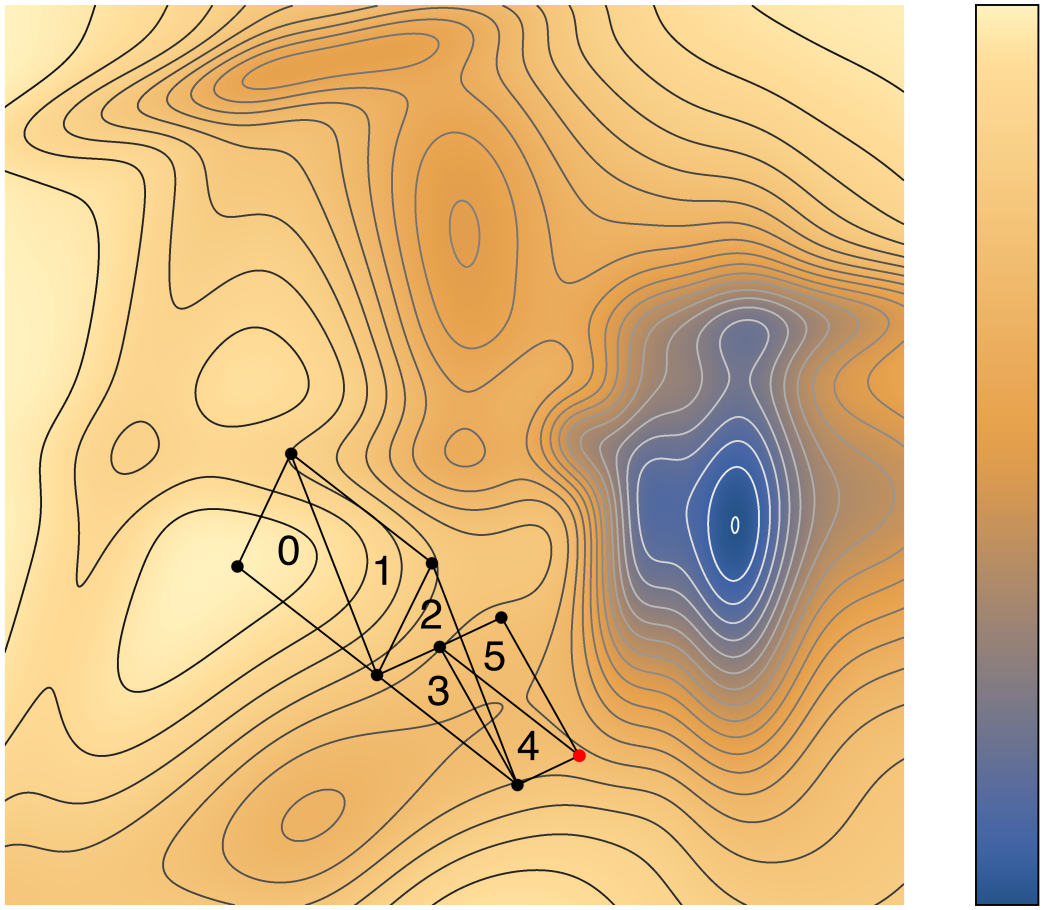

Figure 1: See question 3(d).

by $N-1$ instead of N). Alternatively, we can view this as a Poisson problem. Your first thought may then be to calculate the scatter as $\sigma = \sqrt{N_{\text{inside}}}/N \times A_{\text{box}} = \sqrt{f}/\sqrt{N} \times A_{\text{box}}$, but this will always slightly overestimate the actual scatter by a factor $\sqrt{f}/\sqrt{f-f^2} = 1/\sqrt{1-f}$. This is because there is a bound to the problem here: we cannot get more than N points inside the shaded region, but the Poisson variance assumes the function can be integrated to $N_{\text{inside}} = \infty$. Still, you get half a bonus point for giving the naive Poisson error.

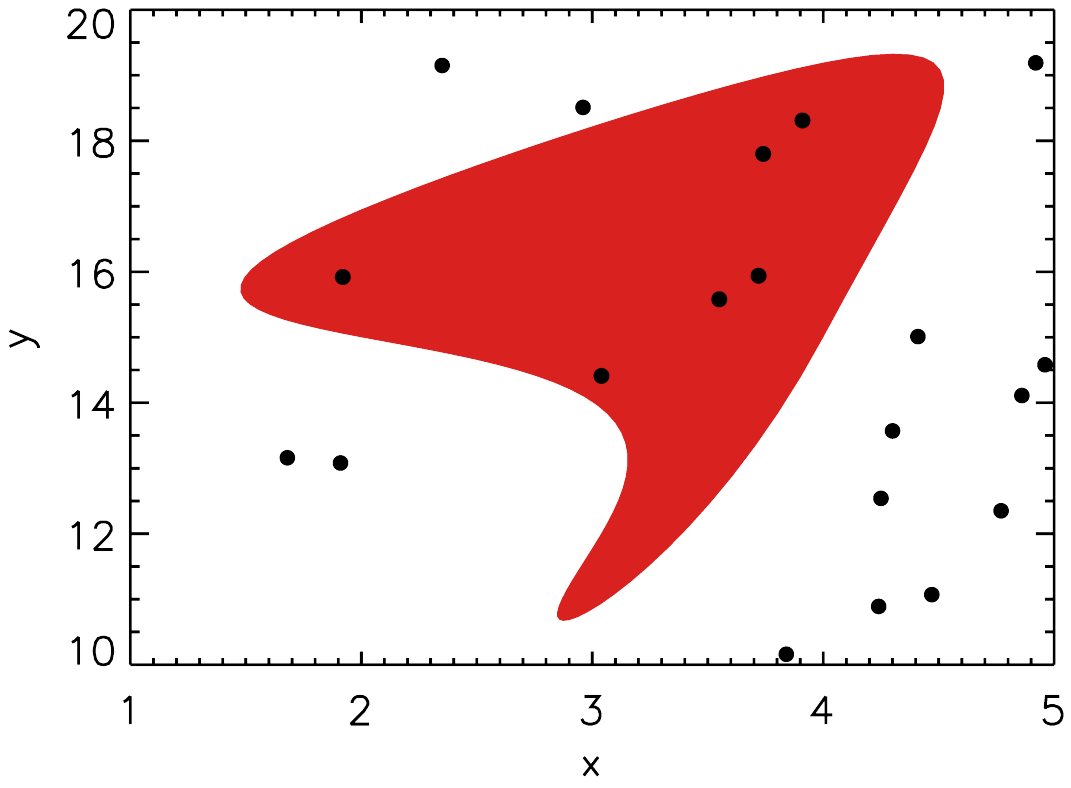

Figure 2: See question 3(e).